

Demonstration of Proposed Features

Date	02 July 2026
Team ID	CCAP-01
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	Data Preprocessing & Feature Engineering	Merge application and credit record datasets by ID, convert repayment status into a binary approval target, clean records, and engineer features such as age, years employed, income category, and financial stability score.	Implemented	Yes	preprocessing.py
2	Multi-Model Training & Comparison	Train and compare eight classification models (Logistic Regression, Decision Tree, Random Forest, XGBoost, SVM, KNN, Naive Bayes, Gradient Boosting) using accuracy, precision, recall, F1, and ROC-AUC.	Implemented	Yes	train.py, reports/model_comparison.csv

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3	Best Model Selection & Persistence	Select the best-performing model and persist it along with metadata for reuse in the prediction pipeline.	Implemented	Yes	models/best_model.pkl, model_metadata.pkl
4	Flask Web App: Single & Batch Prediction	Web application with a home page, single applicant prediction form, and batch CSV prediction, returning Approved/Rejected results with approval probability.	Implemented	Yes	app.py, predict.py, templates/prediction.html, batch_results.html
5	User Accounts, History & Report Downloads	Signup/login, prediction history tracking, and downloading predictions as PDF or Excel, plus model information and feature reference pages.	Implemented	Yes	templates/login.html, signup.html, admin.html, model.html, features.html
6	Testing, Validation & Documentation	Automated tests for validation, credit-rule behavior, downloads, history, and login/signup; input validation, custom error pages, and full project documentation.	Implemented	Yes	tests/test_app.py, templates/404.html, 500.html, README.md

Feature Implementation Summary:

Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100%